SmartHire AI

AI-Powered Resume Screening, Job Recommendation & Skill Gap Analysis System

1. Project Overview

SmartHire AI is an end-to-end classical Machine Learning framework that automates resume parsing, resume classification, content-based job recommendation, career-specific skill-gap analysis and candidate fit assessment. The system integrates supervised learning, unsupervised learning, information retrieval and semantic similarity to assist job seekers in identifying suitable career opportunities and missing competencies.

2. System Workflow

Resume (PDF/DOCX)

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Resume Parser

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Text Cleaning & Preprocessing

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Parallel ML Modules

■■■ TF-IDF + Random Forest → Resume Classification

■■■ TF-IDF + Cosine Similarity → Job Recommendation

■■■ Career Selection → Core Role Aggregation → Required Skills

■■■ Resume Skill Extraction → Exact + Word2Vec Semantic Matching → Skill Gap Report

3. Technical Pipeline

Module	Technique	Output
Resume Parser	PyMuPDF / python-docx	Resume Text
Resume Classification	TF-IDF + Random Forest	Career Category
Job Recommendation	TF-IDF + Cosine Similarity	Top-N Jobs
Skill Aggregation	Core Role + Frequency Analysis	Top Technical Skills
Semantic Matching	Word2Vec	Related Skills
Frontend	Streamlit	Interactive Dashboard

4. Skill Gap Analysis Flow

Candidate Resume → Extract Complete Technical Skill Set

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User Selects Target Career

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Identify Core Role

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Aggregate Skills from Similar Roles

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Rank Top Required Skills (Frequency-Based)

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Exact Skill Matching

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Word2Vec Semantic Matching

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Matched Skills • Related Skills • Missing Skills • Match Score

5. Methodology

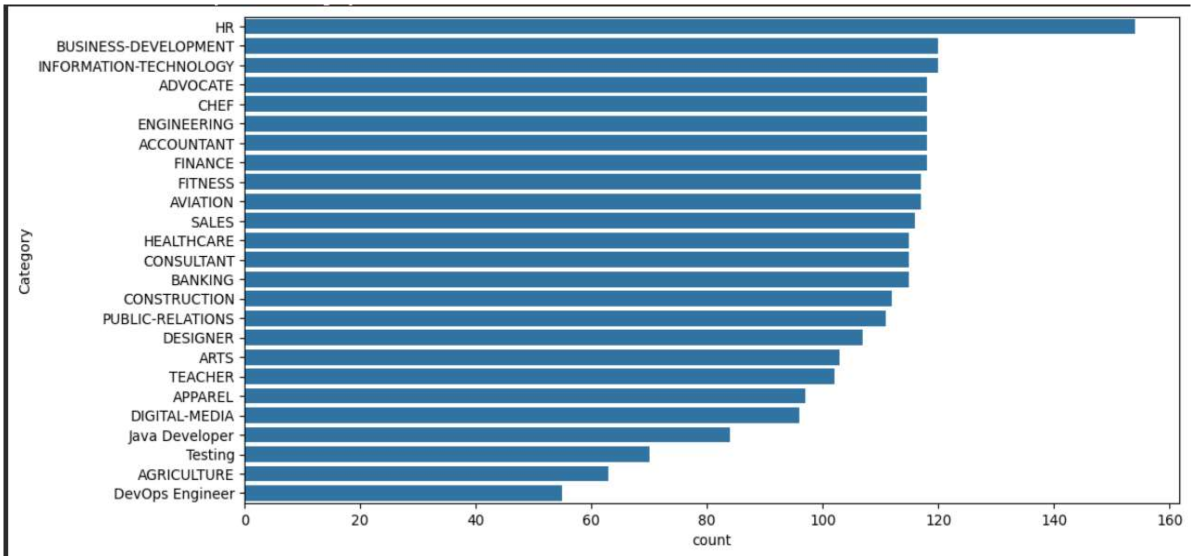
Data preprocessing includes duplicate removal, lowercase normalization, whitespace normalization, skill normalization and title standardization. TF-IDF converts textual resumes and job descriptions into sparse feature vectors. Random Forest predicts resume category. Cosine similarity retrieves the most relevant jobs. K-Means clustering groups similar job profiles to derive career families during preprocessing. A master skill dictionary extracts technical skills from resumes, while Word2Vec identifies semantically related technologies such as TensorFlow–Keras or Pandas–NumPy to improve skill-gap analysis.

6. Key Deliverables

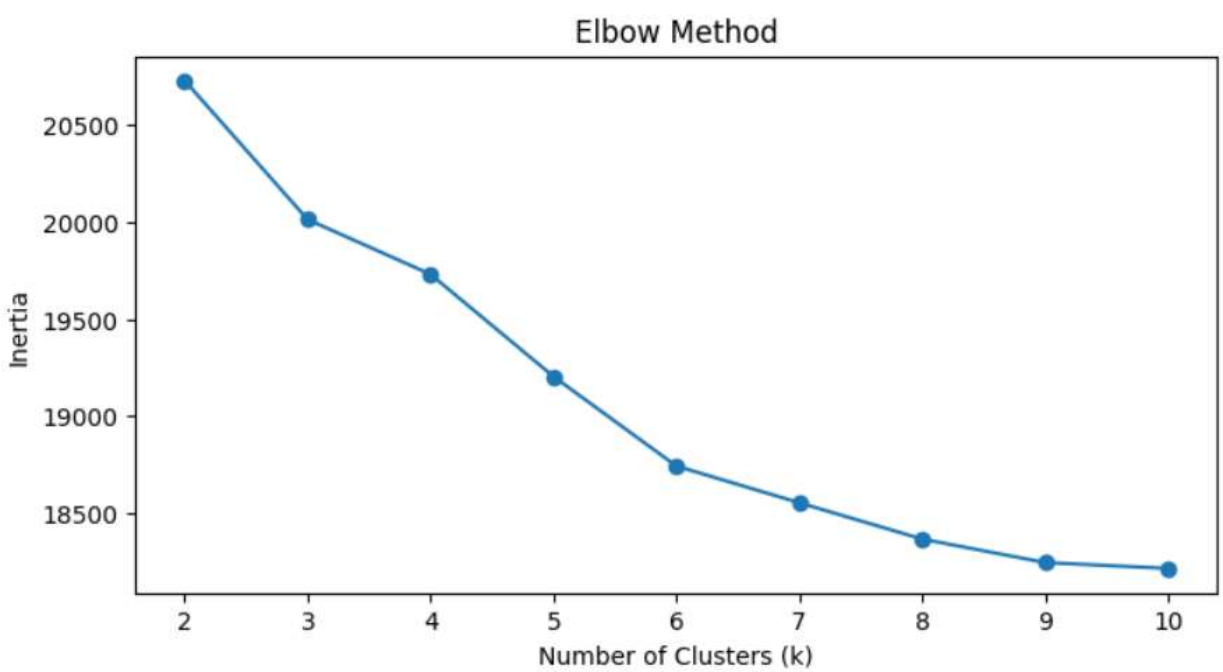
- Resume Category Prediction
- Top-N Personalized Job Recommendations
- Career Search Interface
- Top Required Skills for Selected Career
- Resume Skill Extraction
- Exact and Semantic Skill Matching
- Missing Skill Identification
- Overall Skill Match Score

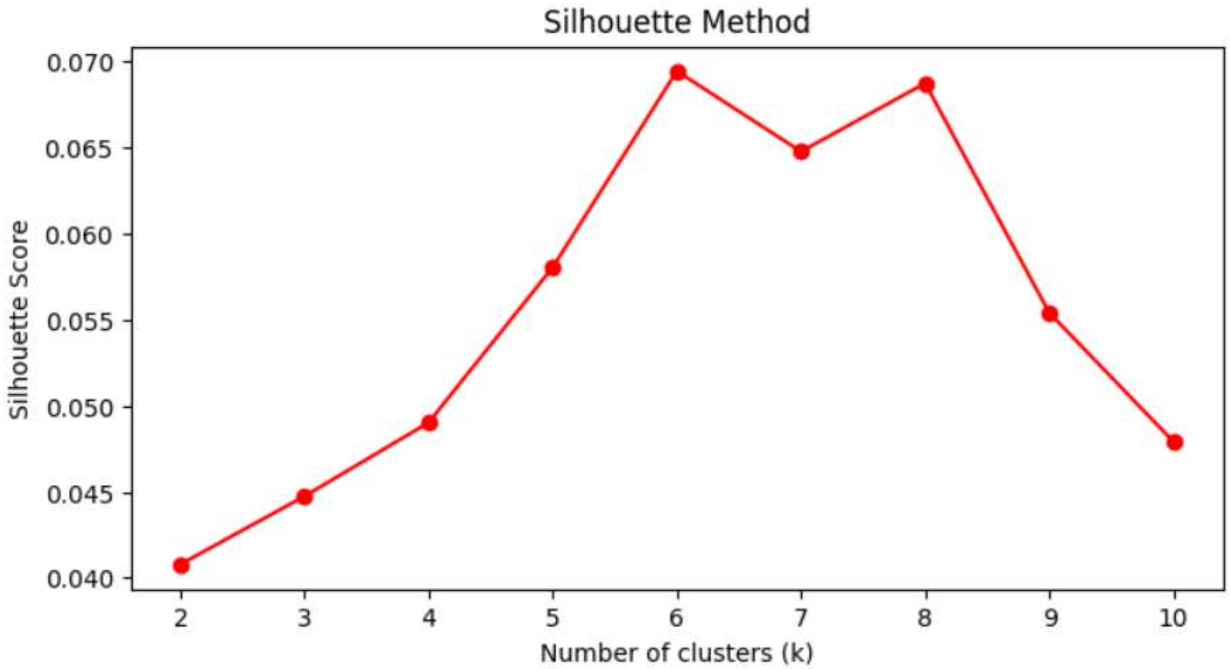
Key Visualization OF ML Models:

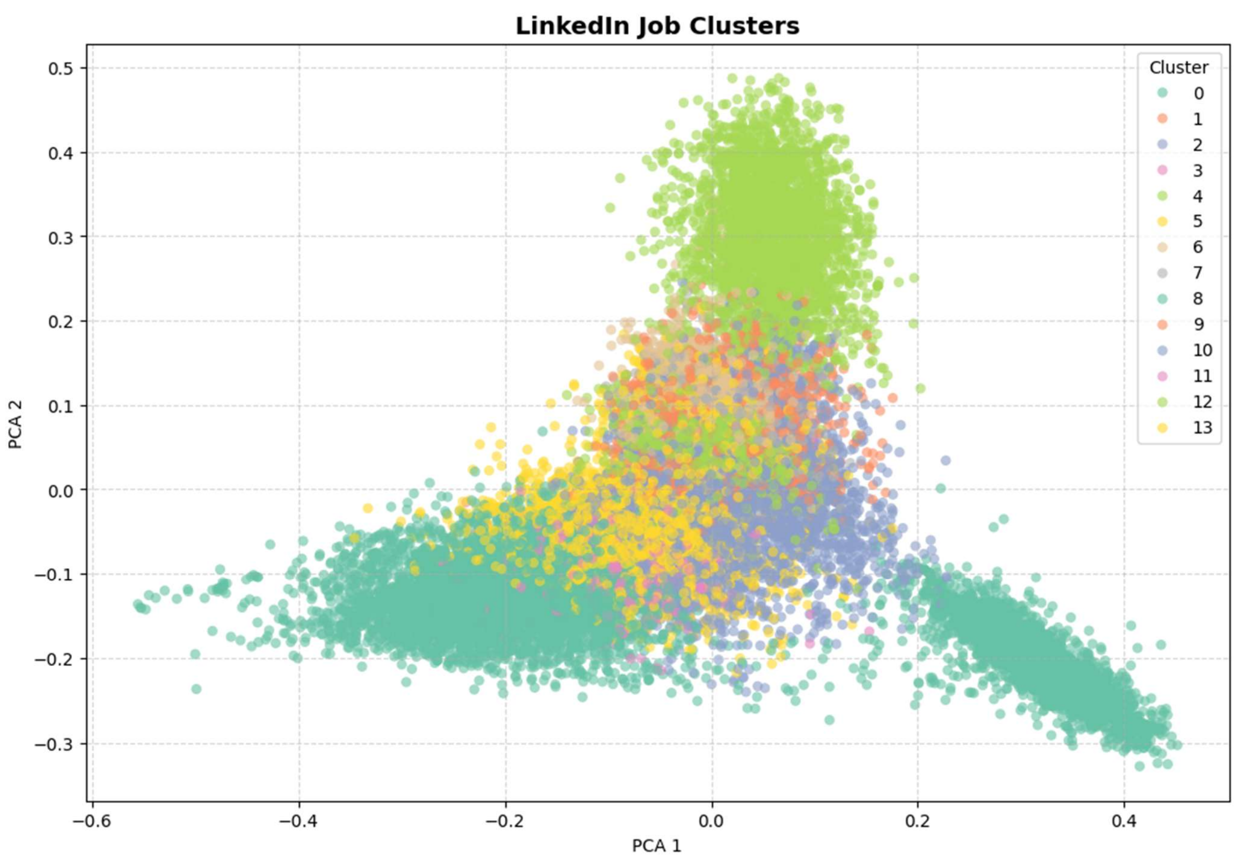
1. **Resume Classification Model**



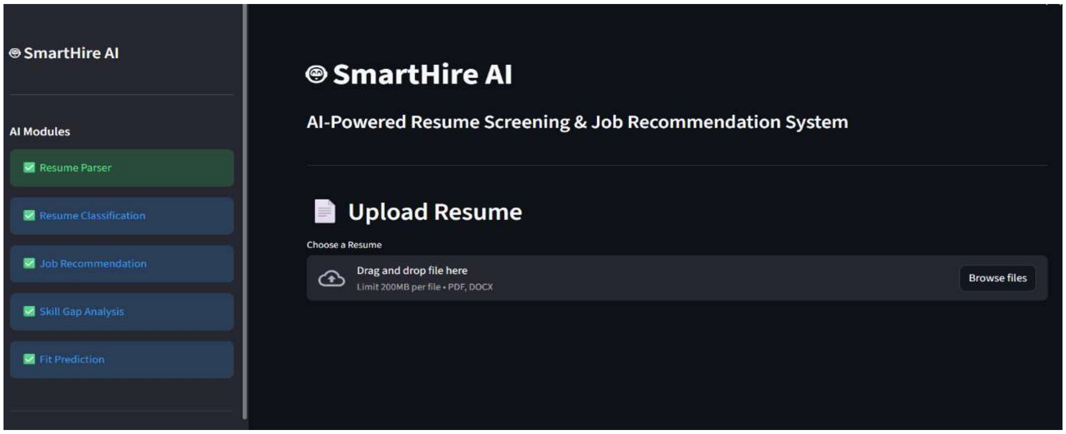
2. **Clustering Model**

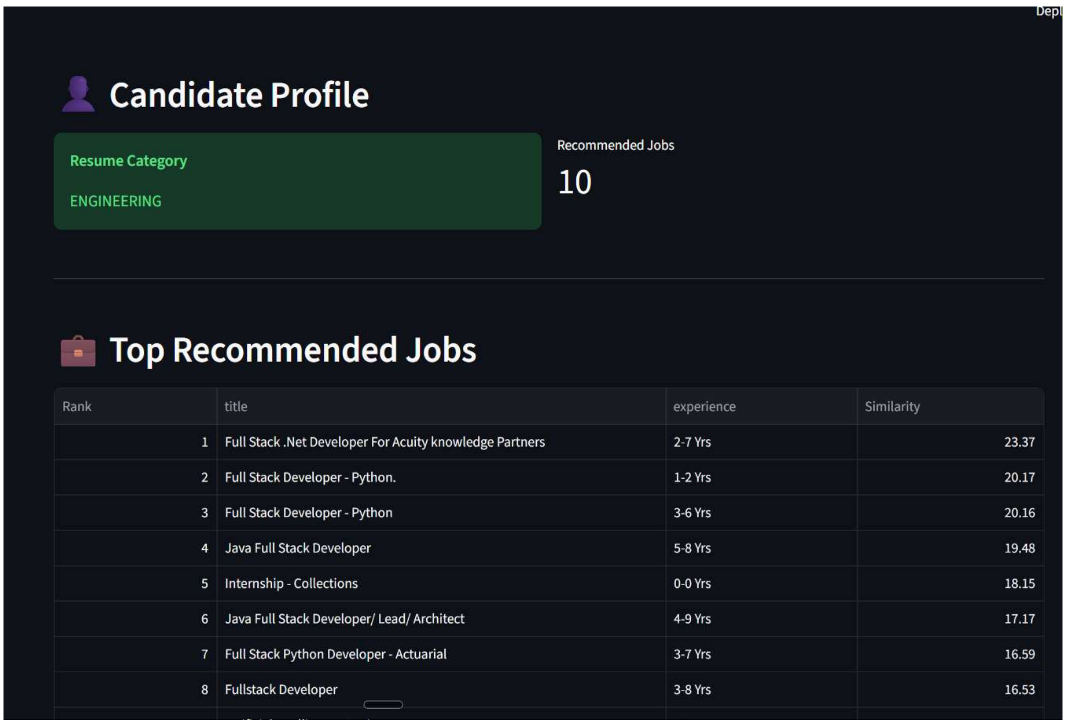


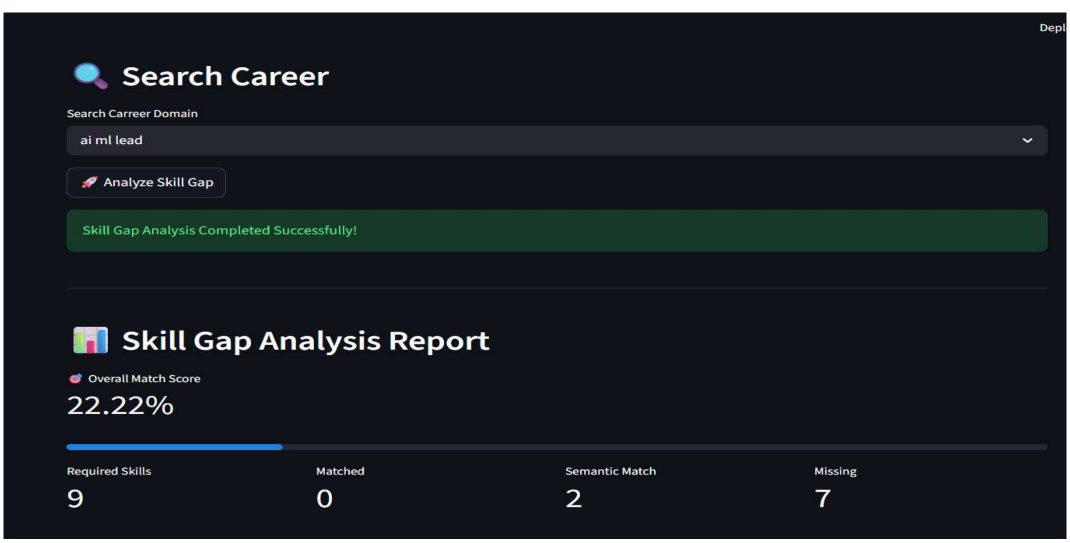

Silhouette Method
Silhouette Score
0.070
0.065
0.060
0.055
0.050
0.045
0.040
2
3
4
5
6
7
8
9
10
Number of clusters (k)


LinkedIn Job Clusters
Cluster
0
1
2
3
4
5
6
7
8
9
10
11
12
13
PCA 2
PCA 1
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.6
−0.4
−0.2
0.0
0.2
0.4

7. RESULT :


SmartHire AI
AI Modules
Resume Parser
Resume Classification
Job Recommendation
Skill Gap Analysis
Fit Prediction
SmartHire AI
AI-Powered Resume Screening & Job Recommendation System
Upload Resume
Choose a Resume
Drag and drop file here
Limit 200MB per file • PDF, DOCX
Browse files


Candidate Profile
Resume Category
ENGINEERING
Recommended Jobs
10
Top Recommended Jobs
Rank title experience Similarity
1 Full Stack .Net Developer For Acuity knowledge Partners 2-7 Yrs 23.37
2 Full Stack Developer - Python. 1-2 Yrs 20.17
3 Full Stack Developer - Python 3-6 Yrs 20.16
4 Java Full Stack Developer 5-8 Yrs 19.48
5 Internship - Collections 0-0 Yrs 18.15
6 Java Full Stack Developer/ Lead/ Architect 4-9 Yrs 17.17
7 Full Stack Python Developer - Actuarial 3-7 Yrs 16.59
8 Fullstack Developer 3-8 Yrs 16.53


Search Career
Search Carreer Domain
ai ml lead
Analyze Skill Gap
Skill Gap Analysis Completed Successfully!
Skill Gap Analysis Report
Overall Match Score
22.22%
Required Skills
9
Matched
0
Semantic Match
2
Missing
7

8. Conclusion

Smart Hire AI demonstrates a modular, explainable and scalable architecture for intelligent resume analysis. By integrating TF-IDF, Random Forest, Cosine Similarity, K-Means preprocessing and Word2Vec semantic embeddings, the system provides transparent career recommendations and actionable skill-gap insights suitable for academic projects and future industrial enhancement.